

Basic Science Department
Course: Mathematics II
Code: Bas 012



Academic year: 2024/2025
Semester: spring
Assoc. Prof. Mohamed Ali

Assignment (1)

Evaluate the following integrals:

(1) $\int -2x^3 + 3\sqrt[3]{x} + x^{-0.5} dx$

(2) $\int (3 - x)^{7/3} + (2x + 3) dx$

$$(3) \int (\sqrt{x} + 3)^2 dx$$

$$(4) \int \frac{x^{-0.5} + x^4 - 9x^2}{x^2} dx$$

$$5) \int e^x - e^{-5x} + e^{5x+3} dx$$

$$(6) \int 2^{3x+2} + 5^{9-x} dx$$

$$(7) \int (2^x + 3e^x)^2 dx$$

$$(8) \int \frac{(e^{2x} + 2e^x + 1)^2}{e^x + 1} dx$$

$$(9) \int \frac{2}{x} - \frac{1}{7-x} dx$$

$$13) \int 2 \sin 2x \cos 2x \, dx$$

$$(14) \int \cos^2 2\theta \, d\theta$$

$$(15) \int \cosh 6x \cosh 2x \, dx$$

$$(16) \int \sin 2x \cos 6x \, dx$$

$$(17) \int \frac{\tanh \theta}{2 \operatorname{sech} \theta + 1} dx$$

$$(18) \int \cosh \theta [\tanh \theta + \operatorname{sech} \theta] d\theta$$

$$(19) \int \sec 3x$$

$$(20) \int \frac{x^{3/9}}{x^{2/3}} + (\sin x^2 \cos x)^2 dx$$

$$(21) \int \frac{e^y}{2^{2y}} + \frac{\sec^2 y \tan y}{\sec^2 y + 1} dy$$

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Assignment (2)

Evaluate the following integrals:

1. $\int \frac{2}{1+x^2} dx$

2. $\int \frac{dx}{x^2 + 4x + 4}$

3. $\int \frac{dx}{x\sqrt{x^2-4}}$

4. $\int \frac{dx}{(x-1)\sqrt{x^2-2x+2}}$

5. $\int \frac{dx}{x^2+4x+8}$

6. $\int \frac{dx}{\sqrt{9 + 4x - x^2}} dx$

7. $\int (\sin 2x + \cos 2x)^2 dx$

8. $\int \sin 3x \cos 4x \, dx$

9. $\int \operatorname{sech} x (\cosh^3 x + \tanh x) \, dx$

10. $\int (2 + \tan x)^2 \, dx$

11. $\int \frac{\cos^2 x}{1 + \sin x} \, dx$

12. $\int \cosh 2x \cosh 5x \, dx$

13. $\int \frac{-\operatorname{csch}^2 y + \cosh y}{\coth y + \sinh y} \, dy$

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Assignment (3)

Evaluate the following integrals:

1. $\int \frac{\log x + \tanh(\log x)}{x} dx$

2. $\int (5 + \tan x)^7 \sec^2 x dx$

3. $\int \frac{e^x dx}{\sqrt{e^{2x} + 1}}$

4. $\int \frac{\sqrt{\sqrt{x} - 1} \, dx}{\sqrt{x}}$

5. $\int \frac{\sec^2(\cot^{-1} x) + \tan(\cot^{-1} x) \, dx}{1 + x^2}$

6. $\int \frac{5^{\sinh^{-1} x} + \sinh^{-1} x \, dx}{\sqrt{1 + x^2}}$

$$7. \int \frac{\csc^2 x}{\sqrt{\cot^3 x}} dx$$

$$8. \int \left[\frac{5^{\ln x}}{x} + \frac{4}{x \sqrt{1 + (\ln x)^2}} \right] dx$$

9. $\int \frac{\sec(\tan^{-1} x) + 8^{\tan^{-1} x} + \tan^{-1} x}{1 + x^2} dx$

10. $\int \frac{\csc^2(\cos^{-1} x) + \tanh(\cos^{-1} x)}{\sqrt{1 - x^2}} dx$

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Assignment (3)

Evaluate the following integrals:

1. $\int \sinh^3 x \, dx$

2. $\int \cos^4 x \, dx$

3. $\int \sinh^4 x \cosh^3 x \, dx$

4. $\int \sin^4 x \cos^5 x \, dx$

5. $\int \sec^4 x \, dx$

6. $\int \cos^5 x \, dx$

7. $\int \sinh^4 x \, dx$

8. $\int \sinh^5 x \cosh^3 x \, dx$

9. $\int \sinh^3 x \sqrt{\cosh x} \, dx$

10. $\int \operatorname{sech}^6 x \, dx$

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Assignment (5)

Evaluate the following integrals:

1. $\int (x^2 + 2x) 8^{5x+1} dx$

2. $\int e^{\sqrt{x}} + x \cosh 2x \, dx$

3. $\int \tan^{-1} x \, dx$

4. $\int \ln(5x + 3) \, dx$

5. $\int x^2 \log 3x \, dx$

6. $\int \frac{\log x}{x^4} \, dx$

7. $\int x^2 \sqrt{(x+1)^3} \, dx$

8. $\int \sinh^{-1} 2x \, dx$

9. $\int \sec^3 x \, dx$

10. $\int \cosh^{-1} 2x \, dx$

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Assignment (6)

Evaluate the following integrals:

1. $\int \frac{x^3 + 1}{x^4 + 4x + 5} dx$

2. $\int \frac{x - 5}{x^2 - 6x + 8} dx$

3. $\int \frac{6x^2 + 1}{x^3 + x} dx$

4. $\int \frac{3x^3 - 9x^2 - 7x - 3}{x^2 - 3x - 4} dx$

5. $\int \frac{x+2}{(x-1)^2(x-4)} dx$

6. $\int \frac{(2e^x+1)e^x}{(e^x-1)^2(e^x-4)} dx$

$$7. \int \frac{(2 \tan x + 1) \sec^2 x}{(\tan x - 1)^2 (\tan x - 4)} dx$$

$$8. \int \frac{6 \sin x \cos^2 x + \sin x}{\cos^3 x + \cos x} dx$$

Assignment (7)

Prove that:

$$\textcircled{1} \int \sin^n x \, dx = \frac{-1}{n} \sin^{n-1} x \cos x + \frac{n-1}{n} \int \sin^{n-2} x \, dx$$

$$\int \cos^n x \, dx = \frac{1}{n} \cos^{n-1} x \sin x + \frac{n-1}{n} \int \cos^{n-2} x \, dx$$

$$\int \sinh^n x \, dx = \frac{1}{n} \sinh^{n-1} x \cosh x - \frac{n-1}{n} \int \sinh^{n-2} x \, dx$$

$$\int \cosh^n x \, dx = \frac{1}{n} \cosh^{n-1} x \sinh x + \frac{n-1}{n} \int \cosh^{n-2} x \, dx$$

$$\textcircled{2} \int (\ln x)^n dx = x(\ln x)^n - n \int (\ln x)^{n-1} dx$$

$$\textcircled{3} \int x^m (\ln x)^n dx = \frac{x^{m+1}}{m+1} (\ln x)^n - \frac{n}{m+1} \int x^m (\ln x)^{n-1} dx$$

$$\int \frac{(\ln x)^n}{x^m} dx = \frac{-1}{(m-1)x^{m+1}} (\ln x)^n + \frac{n}{m-1} \int \frac{(\ln x)^{n-1}}{x^m} dx$$

$$\textcircled{4} \int x^n \sin x \, dx = -x^n \cos x + n x^{n-1} \sin x - n(n-1) \int x^{n-2} \sin x \, dx$$

$$\int x^n \cos x \, dx = x^n \sin x + n x^{n-1} \cos x - n(n-1) \int x^{n-2} \cos x \, dx$$

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Assignment (8)

- 1- Find the area of the region enclosed by the parabola $y = 2 - x^2$ and the line $y = -x$.

2- Find the area of the region enclosed by the curve $y = \sin x$, $y = \cos x$, $x = \frac{\pi}{2}$, and the y - axis.

3- Find the area of the region enclosed by the curve $x = \frac{1}{2}y^2 - 3$, $y = x - 1$.

4- Find the area of the region enclosed by the curve $y = \sec x$, $y = \tan x$, in the interval $0 \leq x \leq \pi/4$.

5- Find the area in the first quadrant enclosed by the curves $y = \sqrt{x - 1}$, $y = 3 - x$, the x -axis, and the y -axis by using definite integrals along the y -axis.